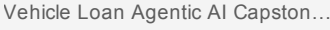


Yes. **AI Financial Crime Investigation Copilot** is an excellent topic for the type of capstone blueprint in your attached example because it has a clear business problem, multiple data sources, ML + GenAI + agentic AI, human decision-making, compliance, auditability, and measurable business outcomes.

It is also highly current. FATF's 2026 work highlights cyber-enabled fraud, increasingly complex cross-border illicit finance, DeFi risks, underground banking/hawala and the need for better information sharing.  FATF +2 McKinsey specifically identifies KYC/AML and financial-crime investigations as a high-potential area for agentic AI.  McKinsey ...

Your sample capstone explicitly expects the participant to map the business process, data sources, AI components, human-in-the-loop controls, guardrails, architecture and implementation roadmap. 

Proposed Capstone Title

Agentic AI Financial Crime Investigation Copilot for Banking

A slightly more leadership-oriented title could be:

Transforming Financial Crime Investigation Using Agentic AI: An AI Copilot for AML Alert Investigation and Decision Support

1. What problem are we solving?

Imagine a bank processes millions of transactions.

Its transaction-monitoring systems identify suspicious activity such as:

₹2 lakh → Account B → Account C → Account D → overseas account

or:

Customer normally transacts ₹50,000–₹1 lakh per month → suddenly receives ₹25 lakh → distributes it across 12 accounts within 24 hours.

The monitoring system generates an **AML alert**.

The problem is that an investigator may then have to manually examine the customer's KYC profile, account history, transaction history, counterparties, previous alerts, sanctions/PEP screening, adverse-media information and internal policies before deciding whether the activity is explainable or warrants escalation.

This creates a strong AI opportunity.

McKinsey notes that banks commonly dedicate substantial staffing to KYC/AML and that fragmented data and low automation leave teams doing significant manual work.

 McKinsey & C...

So your capstone's central problem could be:

How can AI reduce the time required to investigate financial-crime alerts while improving consistency, traceability and investigator decision support without allowing AI to make uncontrolled compliance decisions?

That's a strong **Leadership with AI** problem.

2. Proposed AI Solution

Instead of replacing the investigator, create an:

AI Financial Crime Investigation Copilot

When an AML alert arrives, the Copilot automatically gathers the evidence, analyzes it and prepares an investigation package.

For example:

Alert: Unusual movement of ₹25 lakh

The Copilot investigates:

Customer profile

- Occupation: Salaried
- Expected monthly turnover: ₹1 lakh
- Account age: 6 years
- Risk rating: Medium

Transaction analysis

- ₹25 lakh received
- Funds transferred to 12 accounts
- 90% transferred within 24 hours
- Transaction behaviour significantly different from historical pattern

Counterparty analysis

- 12 beneficiary accounts
- 4 newly added
- 2 associated with previously flagged accounts

KYC / Screening

- KYC valid
- No direct sanctions match
- Relevant adverse-media information, if supported by approved sources

Then it produces something like:

Investigation Summary

The transaction pattern is inconsistent with the customer's historical activity. Rapid movement of funds across multiple newly added beneficiaries requires further investigation.

Recommended action: Escalate for investigator review.

Evidence: transaction IDs, customer profile, counterparties and alert history.

Importantly, **the human investigator remains responsible for the consequential decision.**

That makes your project much stronger from a leadership/governance perspective.

3. Where does AI actually come into the solution?

This is where the project becomes particularly interesting.

Component	AI technique
Suspicious transaction detection	ML / anomaly detection
Transaction pattern recognition	ML
Customer behaviour comparison	ML
Relationship/network identification	Graph analytics
Document analysis	OCR + NLP
Adverse-media analysis	NLP / GenAI
Bank policy lookup	RAG
Investigation summary	GenAI
Case orchestration	Agentic AI

Component	AI technique
Case recommendation	Rules + ML + AI decision support
Final escalation/closure	Human investigator

This is similar to your vehicle-loan blueprint, which deliberately separates rules, ML, OCR, GenAI, agents and human judgment rather than calling everything "GenAI."

Vehicle Loan Agentic AI Capston...

4. Agentic AI Architecture

You could create **7–8 specialized agents**.

1. Alert Triage Agent receives the AML/fraud alert, understands its reason and determines what investigation is required.

↓

2. Customer Intelligence Agent collects KYC, customer risk rating, occupation/business, account history and previous alerts.

↓

3. Transaction Investigation Agent analyzes historical transactions, unusual amounts, frequency, velocity and fund movement.

↓

4. Counterparty & Network Agent identifies relationships among accounts, counterparties and previously flagged entities.

↓

5. Screening & Adverse Media Agent checks approved sanctions/PEP/watchlist sources and approved adverse-media sources.

↓

6. Policy & Compliance Agent uses RAG to retrieve relevant AML policies, investigation procedures and escalation requirements.

↓

7. Investigation Copilot Agent consolidates the evidence and produces an investigator-readable case summary.

↓

8. QA / Validation Agent checks whether the conclusions are actually supported by evidence and whether required investigation steps were completed.

↓

Human Investigator

Reviews:

Evidence → AI analysis → reason codes → recommendation

and makes the authorized decision.

This concept is realistic: McKinsey describes agentic financial-crime architectures using specialized squads for data extraction, ownership analysis, PEP/sanctions screening, transaction analysis, adverse-media screening and consolidated case preparation, with human supervision and audit trails.  McKinsey & C...

5. Example End-to-End Business Process

Your blueprint could have approximately 12 process steps:

P1: AML monitoring system generates alert

P2: AI triages alert

P3: Customer/KYC information retrieved

P4: Historical transactions analyzed

P5: Suspicious transactions identified

P6: Counterparties/network analyzed

P7: Previous alerts/cases analyzed

P8: Sanctions/PEP/adverse-media checks performed

P9: Relevant AML policies retrieved through RAG

P10: AI generates investigation summary and evidence pack

P11: Human investigator reviews and decides/escalates

P12: Case outcome, rationale and audit trail recorded

That is almost ideal for the **Business Process Decomposition** section of your capstone.

6. Data Model

You can create entities similar to the data model in the attached blueprint.

For example:

Customer

customerID, name, occupation, geography, riskRating

Account

accountID, customerID, accountType, openingDate, status

Transaction

transactionID, accountID, amount, date, channel, beneficiary, country

Counterparty

counterpartyID, accountNumber, bank, geography, riskIndicators

AMLAAlert

alertID, scenario, riskScore, triggerReason, timestamp

InvestigationCase

caseID, alertID, investigator, status, disposition

KYCProfile

customerID, KYCStatus, PEPStatus, riskClassification

ScreeningResult

sanctionsMatch, PEPMatch, adverseMediaResult

AllInvestigation

riskIndicators, anomalies, networkFindings, AIRecommendation, confidenceScore

AuditLog

agent, action, source, timestamp, output, humanOverride

That last entity is especially important because your sample blueprint repeatedly emphasizes **auditability and traceability**.

Vehicle Loan Agentic AI Capston...

7. A Good Scenario for Your Presentation

Suppose:

Normal customer behaviour

Salary: ₹1.5 lakh/month

Average transactions: ₹50,000–₹2 lakh

Then suddenly:

₹40 lakh received

Within 6 hours:

₹8 lakh → Account A

₹7 lakh → Account B

₹6 lakh → Account C

₹5 lakh → Account D

₹7 lakh → Account E

₹7 lakh → Account F

The AI system discovers:

- Sudden deviation from historical behaviour
- Multiple new beneficiaries
- Rapid movement of funds
- Counterparty links to previous alerts
- Potential adverse-media information

The Copilot creates a **transaction timeline + network map + investigation summary + evidence references**.

The investigator doesn't have to manually assemble everything.

This is the story I'd use to explain the project to your professor because the problem and AI value become immediately understandable.

8. Leadership Angle — the most important part

Don't present the capstone as:

"I am creating an AI model to detect money laundering."

That becomes primarily a data-science project.

Instead present it as:

"I am proposing an AI-enabled transformation of the financial-crime investigation

operating model."

Leadership has to decide:

What should AI automate?

Data gathering, transaction analysis, summarization, policy retrieval and case preparation.

What should AI recommend?

Risk indicators, investigation priorities and escalation suggestions.

What should humans control?

Consequential case decisions, regulatory escalation, exceptions and overrides.

How should AI be governed?

Explainability, source attribution, access controls, model monitoring, privacy, audit trails and human oversight.

This aligns very closely with your sample capstone's philosophy: the participant acts as an **architect/AI leader**, defining business, data, AI, technology, governance and execution blueprints rather than merely building a model.

Vehicle Loan Agentic AI Capston...

Why this is a particularly good 2026 topic

Financial crime is changing quickly. FATF reported in February 2026 that **90% of assessed jurisdictions identified fraud as a major money-laundering risk**, while its September 2026 work highlights professional money laundering through underground banking and hawala-like networks.  FATF is also emphasizing stronger public-private information sharing as transactions become more digital and cross-border. 

So the project gives you a very credible story:

Current problem → fragmented/manual investigations → AI-enabled investigation → agentic orchestration → human decision → governance → measurable business impact.

And it gives you plenty of material for essentially **every section of the 14-page sample blueprint** you attached: data model, process matrix, agentic workflow, UML diagrams, component architecture, build/buy/partner decisions, guardrails, HITL authority matrix, implementation roadmap and KPIs.

Vehicle Loan Agentic AI Capston...

If you choose this topic, the next useful step would be to define the **exact capstone problem statement, scope, executive summary and current-state vs future-state**

process before developing the full blueprint.

Sources

Is this conversation helpful so far?